Chapter 6. Training and Optimization of Deep Networks

Mini-batch SGD, activations, weight initialization, learning rates, batch normalization; Taylor expansion and the descent lemma, SGD, momentum, Nesterov, AdaGrad, RMSProp, Adam

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1 Full, Stochastic and Mini-Batch Gradients

Motivation. Training minimizes an average loss over millions of examples, and computing the exact gradient at each step is too expensive. Gradients on small random batches are unbiased and cheap, which is the basis of all practical training.

For $f(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N} f_i(\boldsymbol{\theta})$, the full gradient costs N per-sample gradients. Choosing I uniformly, $\mathbf{g} = \nabla f_I$ is **unbiased**: $\mathbb{E}[\mathbf{g} \mid \boldsymbol{\theta}] = \frac{1}{N}\sum_i \nabla f_i = \nabla f$. A mini-batch averages b such gradients.

Theorem 6.1: mini-batch variance

If the b indices are drawn independently with replacement and a single draw has conditional covariance Σ, the mini-batch gradient has covariance Σ/b. Without replacement it is $\frac{N-b}{b(N-1)}\Sigma$ (with Σ normalized by N).

Proof. With replacement the b terms are iid with covariance Σ; the covariance of their average is $\frac{1}{b^2}\cdot b\Sigma$. Without replacement the pairwise covariances are $-\Sigma/(N-1)$, which gives the finite-population correction. ■

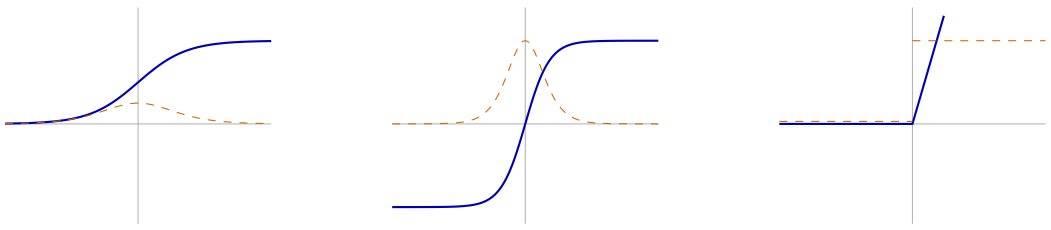
Example 6.2: unbiased but noisy

$f_i(w) = \frac{1}{2}(w - y_i)^2$ with $\mathbf{y} = (1, 3, 5)$ at $w = 0$: per-sample gradients $-1, -3, -5$, full gradient -3. A single random gradient averages to -3; a two-sample batch without replacement gives $-2, -3, -4$, again averaging to -3.

One *iteration* is one parameter update; one *epoch* is one pass through the data (1000 samples, batch 100: 10 iterations per epoch). Full batch: deterministic but expensive; single sample: cheap but noisy; mini-batch: a compromise.

2 Activation Functions Revisited

Sigmoid. Saturates: $\sigma' \le 1/4$ and $\sigma' \to 0$ for large $|z|$. Outputs are positive, so for the next layer $h = W\mathbf{z}$ (with $\mathbf{z}$ the positive sigmoid outputs) $\partial L/\partial W_{ij} = \delta_i z_j$ has the sign of δ_i for every j: all weights into a neuron move in the same direction in one step, producing zig-zag paths. **Tanh.** Zero-centered and odd, $\tanh' \in (0, 1]$, but still saturates near ± 1. **ReLU.** Cheap, derivative 1 on the positive side (no saturation there), but units with negative input get zero gradient (dead ReLU) and outputs are not centered. Leaky ReLU, ELU and maxout keep a nonzero slope or add flexibility.



Activation functions (solid) and their derivatives (dashed): the sigmoid and tanh *saturate, ReLU has derivative* 0 *or* 1.

3 Weight Initialization

3.1 Why the scale matters

`W = s * np.random.randn(Din, Dout)` draws iid $N(0, s^2)$ weights. For $z_j = \sum_{i=1}^{D_{\text{in}}} W_{ji} x_i$ with zero-mean independent weights of variance σ_W^2, independent of inputs with $\mathbb{E}x_i^2 = q$,

$$\mathbb{E}z_j = 0, \qquad \operatorname{Var} z_j = D_{\text{in}} \sigma_W^2 q$$

(cross terms vanish because $\mathbb{E}[W_{ji}W_{jk}] = 0$ for $i \neq k$). For six tanh layers of width 4096:

- $s = 0.01$: $D_{\text{in}} s^2 = 0.41$; activations shrink towards 0 layer by layer, and weight gradients $\propto$ inputs become tiny.
- $s = 0.05$: $D_{\text{in}} s^2 = 10.24$; pre-activations have standard deviation ≈ 3, tanh saturates (histograms pile up at ± 1), $\tanh' \approx 0$, and the backpropagated signal $\partial L/\partial \mathbf{z} = \sigma'(\cdot) W^\top \partial L/\partial \mathbf{h}$ is almost zero: no learning, although the loss is still large. A zero gradient caused by saturation is not a sign of optimality.

3.2 Zero and symmetric initialization

Theorem 6.3: zero initialization freezes a tanh MLP

For $f(\mathbf{x}) = A_L y_{L-1} + b_L$, $y_\ell = \tanh(A_\ell y_{\ell-1} + b_\ell)$, all parameters initialized to 0 and GD on $\frac{1}{2}\sum_i (f(X_i) - Y_i)^2$: $A_1, \ldots, A_L, b_1, \ldots, b_{L-1}$ stay 0 forever, and $b_L \to \bar{Y}$ if $0 < \eta < 2/N$.

Proof. Induction on the claim "all parameters except b_L are zero". Forward: $\tanh(0) = 0$ so every hidden output is 0 and $f = b_L$. Backward: $\partial/\partial A_L = \sum_i \delta_L^{(i)} y_{L-1}^\top = 0$; $\boldsymbol{\delta}_{L-1} = (A_L^\top \delta_L) \odot \tanh'(0) = 0$ since $A_L = 0$, and inductively all $\boldsymbol{\delta}_\ell = 0$, so $\partial A_\ell = \boldsymbol{\delta}_\ell y_{\ell-1}^\top = 0$, $\partial b_\ell = 0$. Hence GD does not move them. For b_L: $J = \frac{1}{2}\sum (b_L - Y_i)^2$, $b^{(t+1)} - \bar{Y} = (1 - \eta N)(b^{(t)} - \bar{Y})$. ■

Theorem 6.4: duplicate neurons stay duplicates

For $f(x) = \sum_{j=1}^{p} u_j \sigma(a_j x + b_j)$ trained by Adam (or GD, momentum, RMSProp), if $(a_{p-1}, b_{p-1}, u_{p-1}) = (a_p, b_p, u_p)$ at initialization, they remain equal for all iterations.

Proof. $\partial f/\partial u_j = \sigma(a_j x + b_j)$, $\partial f/\partial a_j = u_j \sigma'(a_j x + b_j) x$, $\partial f/\partial b_j = u_j \sigma'(a_j x + b_j)$; the loss gradient multiplies these by $\partial \ell/\partial f$, which is shared by all neurons. So equal parameters give equal gradient coordinates. Induct on the statement "the paired coordinates of $\boldsymbol{\theta}^k$, m_1^k, m_2^k agree": it holds at $k = 0$ ($m_1^0 = m_2^0 = 0$), and every Adam operation (EMAs, bias corrections, elementwise division) acts coordinatewise. The optimizer state must be part of the induction hypothesis. ■

Hence identical neurons compute the same function forever: random initialization is needed to break symmetry.

3.3 Xavier and He initialization

For an approximately linear activation, preserving the second moment requires $D_{\text{in}} \sigma_W^2 = 1$, i.e. $\sigma_W = 1/\sqrt{D_{\text{in}}}$ (often called "Xavier"; Glorot's compromise $2/(D_{\text{in}} + D_{\text{out}})$ also balances the backward pass).

Theorem 6.5: He initialization

Let $y^+ = A\mathbf{x}$, $\mathbf{x} = \mathrm{ReLU}(\mathbf{y})$, $A_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_A^2)$, where $\mathbf{y}$ is independent of A, its coordinates have mean 0, variance 1, are uncorrelated, and $\mathbf{y} \overset{d}{=} -\mathbf{y}$. Then

(i) $\mathbb{E}x_j^2 = \frac{1}{2}$; (ii) $\mathbb{E}y^+ = 0$; (iii) $\mathbb{E}(y_i^+)^2 = \frac{n_{\text{in}}\sigma_A^2}{2}$;

(iv) $y_1^+, \dots, y_{n_{\text{out}}}^+$ are uncorrelated; (v) $y^+ \overset{d}{=} -y^+$.

Thus $\sigma_A^2 = 2/n_{\text{in}}$ preserves the second moment through ReLU layers.

Proof. (i) $\mathrm{ReLU}(y)^2 = y^2\mathbf{1}\{y > 0\}$; symmetry makes the positive and negative halves of $\mathbb{E}y^2 = 1$ equal ($y = 0$ contributes nothing).

(ii) Independence and $\mathbb{E}A_{ij} = 0$.

(iii) $\mathbb{E}(y_i^+)^2 = \sum_{j,k} \mathbb{E}[A_{ij}A_{ik}]\mathbb{E}[x_j x_k] = \sum_j \sigma_A^2 \mathbb{E}x_j^2$.

(iv) For $i \neq k$ all $\mathbb{E}[A_{ij}A_{kl}] = 0$.

(v) $(A, \mathbf{y}) \overset{d}{=} (-A, \mathbf{y})$ since A is independent of $\mathbf{y}$ and symmetric; apply $F(A, \mathbf{y}) = A\,\mathrm{ReLU}(\mathbf{y})$. ■

Remark 6.6: refinements

Leaky ReLU with slope α: $\mathbb{E}\phi(y)^2 = \frac{1+\alpha^2}{2}$ and $\sigma_A^2 = \frac{2}{(1+\alpha^2)n_{\text{in}}}$. ReLU output is not centered: for $z \sim N(0, v)$, $\mathbb{E}\,\mathrm{ReLU}(z) = \sqrt{v/(2\pi)}$, $\mathbb{E}\,\mathrm{ReLU}(z)^2 = v/2$, so He preserves the second moment, not the variance. Backward, $\partial\ell/\partial\mathbf{x} = A^\top \partial\ell/\partial\mathbf{y}^+$ has variance $n_{\text{out}}\sigma_A^2$: forward and backward preservation ask for n_{in} versus n_{out}.

Theorem 6.7: sigmoid layer

For $\mathbf{y} = \sigma(A\mathbf{x})$ with $\sigma(t) \approx \frac{1}{2} + \frac{t}{4}$, $\sigma' \approx \frac{1}{4}$, $A_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_A^2)$ independent of $\mathbf{x}$, and x_j uncorrelated with mean $\frac{1}{2}$ and variance 1: $\mathbb{E}y_i = \frac{1}{2}$, $\mathrm{Var}\, y_i = \frac{5n_{\text{in}}\sigma_A^2}{64}$. If the upstream gradients $\partial\ell/\partial y_i$ have mean 0, variance 1, are uncorrelated and independent of A, then $\partial\ell/\partial x_j \approx \frac{1}{4}\sum_i A_{ij}\partial\ell/\partial y_i$ has mean 0 and variance $\frac{n_{\text{out}}\sigma_A^2}{16}$.

Proof. $\mathrm{Var}(A_{ij}x_j) = \mathbb{E}A_{ij}^2\mathbb{E}x_j^2 = \sigma_A^2(1 + \frac{1}{4})$ and the terms are uncorrelated; multiply by $\frac{1}{16}$. The backward computation is identical with $\mathbb{E}(\partial\ell/\partial y_i)^2 = 1$. ■

Theorem 6.8: convolution layer

For a 3×3, padding-1 convolution $Y_{cij} = \sum_{p,q,r} w_{cpqr}\tilde{X}_{p,i+q-2,j+r-2} + b_c$ with zero-mean unit-variance uncorrelated inputs, $w \overset{\text{iid}}{\sim} N(0, \sigma_w^2)$, $b \overset{\text{iid}}{\sim} N(0, \sigma_b^2)$ independent: $\mathbb{E}Y_{cij} = 0$ everywhere, and $\mathbb{E}Y_{cij}^2 = \sigma_w^2 \cdot \#\{\text{in-range inputs}\} + \sigma_b^2$, i.e. $9C_{\text{in}}\sigma_w^2 + \sigma_b^2$ in the interior, $6C_{\text{in}}\sigma_w^2 + \sigma_b^2$ on edges and $4C_{\text{in}}\sigma_w^2 + \sigma_b^2$ at corners.

Proof. Expand the square: weight–weight terms leave $\sigma_w^2\mathbb{E}\tilde{X}_a^2$ on the diagonal, weight–bias terms vanish, and $\mathbb{E}b_c^2 = \sigma_b^2$. Padded entries are zero. ■

3.4 Variance of a sigmoid output

For smooth f and X with mean μ, Taylor's theorem $f(X) = f(\mu) + f'(\mu)(X - \mu) + \frac{1}{2}f''(\xi)(X - \mu)^2$ gives $\mathrm{Var}\, f(X) = f'(\mu)^2\,\mathrm{Var}\, X + \text{Error}$, the error collecting terms with f''. For $X \sim N(0, 1)$

and $Y = \sigma(wX + b)$, $\operatorname{Var} Y \approx w^2\sigma'(b)^2$: small weights or large $|b|$ (saturation) give small output variance. An exact bound also holds: if f is L-Lipschitz, $\operatorname{Var} f(X) \le \mathbb{E}(f(X) - f(\mu))^2 \le L^2 \operatorname{Var} X$, so $\operatorname{Var} Y \le w^2/16$.

4 Learning-Rate Schedules

The learning rate controls the update size: too large diverges, too small is slow, and a large early rate followed by a smaller late rate refines the solution and reduces gradient noise.

- Step: multiply by 0.1 at fixed epochs (ResNet: 30, 60, 90), giving $\eta_0, 0.1\eta_0, 0.01\eta_0, 0.001\eta_0$.
- Cosine: $\eta_t = \frac{\eta_0}{2}[1 + \cos(\pi t/T)]$; linear: $\eta_t = \eta_0(1 - t/T)$; inverse square root: $\eta_t = \eta_0/\sqrt{t}$.
- Linear warmup: increase from 0 to $\eta_{\max}$ over the first iterations (e.g. 5000), then decay.

5 Batch Normalization

5.1 Training-time computation

For a mini-batch $X \in \mathbb{R}^{B\times D}$ (rows: samples, columns: features),

$$\mu_j = \frac{1}{B}\sum_i X_{ij}, \quad v_j = \frac{1}{B}\sum_i (X_{ij} - \mu_j)^2, \quad \hat{X}_{ij} = \frac{X_{ij} - \mu_j}{\sqrt{v_j + \epsilon}}, \quad Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j.$$

γ, β are learned, so the network can choose the output scale and center (with $\gamma_j = \sqrt{v_j + \epsilon}$, $\beta_j = \mu_j$ it recovers the input for that batch).

Example 6.9: two features

$X = \begin{pmatrix} 1 & 10 \\ 3 & 14 \end{pmatrix}$: $\mu = (2, 12)$, $v = (1, 4)$, $\hat{X} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$ (ϵ ignored). With $\gamma = (2, 3)$, $\beta = (-1, 5)$, $Y = \begin{pmatrix} -3 & 2 \\ 1 & 8 \end{pmatrix}$.

Theorem 6.10: invariances

With $\epsilon = 0$:
(i) each column of $\hat{X}$ has mean 0 and mean square 1;
(ii) for $z = \mathbf{w}^\top \mathbf{x} + b$ the BN output does not depend on b and is invariant under $\mathbf{w} \mapsto c\mathbf{w}$, $c > 0$;
(iii) consequently $\langle \nabla_{\mathbf{w}} L, \mathbf{w} \rangle = 0$.

Proof. (ii) $z - \mu = \mathbf{w}^\top(\mathbf{x} - \bar{\mathbf{x}})$ removes b; scaling $\mathbf{w}$ scales both $z - \mu$ and $\sqrt{v}$ by c.
(iii) $h(c) = L(c\mathbf{w})$ is constant, so $0 = h'(1) = \nabla L(\mathbf{w})^\top \mathbf{w}$. ■

5.2 Test time

At inference the batch may have size one, and the output must not depend on the other samples. During training BN keeps running averages

$$\mu^{\text{run}} \leftarrow m\,\mu^{\text{run}} + (1 - m)\mu_{\text{batch}}, \qquad (\sigma^2)^{\text{run}} \leftarrow m\,(\sigma^2)^{\text{run}} + (1 - m)\sigma^2_{\text{batch}}$$

(an exponential moving average, as in Adam; PyTorch's `momentum` argument is $1-m$), and at test time uses

$$\hat{x}_{ij} = \frac{x_{ij} - \mu_j^{\text{run}}}{\sqrt{(\sigma_j^2)^{\text{run}} + \epsilon}}, \qquad y_{ij} = \gamma_j \hat{x}_{ij} + \beta_j.$$

With a batch of one and batch statistics, $\mu = x$ and $v = 0$ would give $\hat{x} = 0$: all information lost. In code: `model.train()` versus `model.eval()`.

Theorem 6.11: fusion at inference

At test time BN is the affine map $z \mapsto az + c$ with $a_j = \gamma_j / \sqrt{v_j^{\text{run}} + \epsilon}$, $c_j = \beta_j - a_j \mu_j^{\text{run}}$, so $\text{BN}(W\mathbf{x} + \mathbf{b}) = \text{diag}(\mathbf{a})W\mathbf{x} + (\text{diag}(\mathbf{a})\mathbf{b} + \mathbf{c})$ can be folded into the preceding layer.

Remark 6.12: why BN helps

Reported benefits: easier optimization, better gradient flow, larger learning rates, less sensitivity to initialization, mild regularization through batch noise. The original "internal covariate shift" explanation has been challenged; a smoother loss landscape is an alternative view. In training mode BN couples samples in the backward pass: with $g_i = \partial L / \partial y_i$ and $r = \sqrt{v + \epsilon}$, $\partial L / \partial x_i = \frac{\gamma}{r}(g_i - \bar{g} - \hat{x}_i \, \overline{g\hat{x}})$.

6 Dropout

Inverted dropout multiplies each unit by $m_i/(1-p)$ with $m_i \sim \text{Bernoulli}(1-p)$ during training (so $\mathbb{E}$ = identity) and does nothing at evaluation. It prevents co-adaptation and acts like training an ensemble of sub-networks (AlexNet applies it to the FC layers).

Theorem 6.13: dropout and activations commute iff positively homogeneous

With the same mask, $\phi(H \odot \mathbf{z}) = H \odot \phi(\mathbf{z})$ for all masks iff $\phi(0) = 0$ and $\phi(cz) = c\phi(z)$ for $c = 1/(1-p)$. This holds for ReLU and LeakyReLU, but fails for the sigmoid.

Proof. Coordinatewise, a dropped unit gives $\phi(0)$ versus 0, a kept unit $\phi(cz)$ versus $c\phi(z)$. For $c > 0$, ReLU and LeakyReLU preserve sign cases and scale linearly. For the sigmoid with $p = 0.4$ and $z = 0$: $\sigma(0)/0.6 \approx 0.833 \neq \sigma(0) = 0.5$. ■

7 Gradient Descent and Taylor's Theorem

Motivation. Why does stepping against the gradient decrease the loss? A first-order Taylor expansion says that for a small enough step it must; the rest of the chapter quantifies "small enough".

We minimize $f : \mathbb{R}^d \to \mathbb{R}$; the iterate $\mathbf{x}_t$ is the whole parameter vector. GD is $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$, and $-\nabla f$ is the steepest-descent direction since $\nabla f^\top \mathbf{v} \geq -\|\nabla f\|$ for unit $\mathbf{v}$ (Cauchy–Schwarz).

Theorem 6.14: Taylor along a segment

With $\mathbf{d} = \mathbf{y} - \mathbf{x}$ and $h(t) = f(\mathbf{x} + t\mathbf{d})$,

$$f(\mathbf{y}) - f(\mathbf{x}) = \int_0^1 \nabla f(\mathbf{x}+t\mathbf{d})^\top \mathbf{d} \, dt, \qquad f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \int_0^1 (1-t) \, \mathbf{d}^\top \nabla^2 f(\mathbf{x}+t\mathbf{d}) \mathbf{d} \, dt.$$

Proof. The first is $h(1) - h(0) = \int_0^1 h'$. For the second, $h(1) = h(0) + h'(0) + \int_0^1 \int_0^t h''(s)\, ds\, dt$ and exchanging the order over $0 \le s \le t \le 1$ gives $\int_0^1 (1-s)h''(s)\, ds$. ■

8 Smoothness and the Descent Lemma

Motivation. If the gradient is L-Lipschitz, the loss lies below a quadratic around each point. This descent lemma gives a safe step size $1/L$ and a guaranteed decrease per step, the standard starting point of convergence proofs.

Definition 6.15: L-smoothness

f is L-smooth if $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \le L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$. For $f \in C^2$ this is equivalent to $-L\mathbf{I} \preceq \nabla^2 f \preceq L\mathbf{I}$ (all Hessian eigenvalues in $[-L, L]$).

Theorem 6.16: descent lemma

If f is L-smooth, $f(\mathbf{y}) \le f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2$.

Proof. Subtract $\nabla f(\mathbf{x})^\top \mathbf{d} = \int_0^1 \nabla f(\mathbf{x})^\top \mathbf{d}\, dt$ from the first Taylor formula, then use Cauchy–Schwarz and smoothness:

$$f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{d} = \int_0^1 (\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x}))^\top \mathbf{d}\, dt \le \int_0^1 Lt\|\mathbf{d}\|^2 dt = \frac{L}{2}\|\mathbf{d}\|^2.$$

■

Theorem 6.17: GD on smooth functions

GD satisfies $f(\mathbf{x}_{t+1}) \le f(\mathbf{x}_t) - \eta(1 - \frac{L\eta}{2})\|\nabla f(\mathbf{x}_t)\|^2$, so it decreases f when $0 < \eta < 2/L$. With $\eta = 1/L$ and $f^\star = \inf f > -\infty$,

$$\min_{0 \le t < T} \|\nabla f(\mathbf{x}_t)\|^2 \le \frac{1}{T} \sum_{t<T} \|\nabla f(\mathbf{x}_t)\|^2 \le \frac{2L\,(f(\mathbf{x}_0) - f^\star)}{T}.$$

Proof. Put $\mathbf{y} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$ in the descent lemma. With $\eta = 1/L$ the coefficient is $1/(2L)$; summing over t telescopes the left side to $f(\mathbf{x}_0) - f(\mathbf{x}_T) \le f(\mathbf{x}_0) - f^\star$. ■

This is a stationarity guarantee: for nonconvex f a small gradient may sit at a saddle point or a local minimum. For $f(x) = x^2/2$ ($L = 1$), $x_{t+1} = (1 - \eta)x_t$ converges iff $0 < \eta < 2$; for $f = \frac{1}{2}\mathbf{x}^\top H\mathbf{x}$ each eigen-direction contracts by $1 - \eta\lambda_i$, so a small eigenvalue (poor conditioning) means slow progress while $\eta < 2/\lambda_{\max}$ is required for stability.

9 Stochastic Gradient Descent

Motivation. With noisy gradients the iterates no longer settle down at a fixed step size. The analysis shows how the noise floor depends on the step size and why decreasing step sizes are needed for convergence.

SGD uses $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \mathbf{g}_t$ with $\mathbb{E}_t \mathbf{g}_t = \nabla f(\mathbf{x}_t)$, where $\mathbb{E}_t$ conditions on the history up to $\mathbf{x}_t$.

Theorem 6.18: second moment is not variance

$\mathbb{E}_t\|\mathbf{g}_t\|^2 = \|\nabla f(\mathbf{x}_t)\|^2 + \mathbb{E}_t\|\mathbf{g}_t - \nabla f(\mathbf{x}_t)\|^2$.

Proof. Expand $\mathbf{g}_t = \nabla f + (\mathbf{g}_t - \nabla f)$; the cross term has conditional mean zero. ■

Theorem 6.19: $T^{-1/2}$ rate for SGD

If f is L-smooth, $\mathbb{E}_t\|\mathbf{g}_t\|^2 \le G$ and $\Delta = f(\mathbf{x}_0) - f^\star$, then for a constant step

$$\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{x}_t)\|^2 \le \frac{\Delta}{\eta T} + \frac{L\eta G}{2}, \qquad \text{and with } \eta = \sqrt{\tfrac{2\Delta}{LGT}}: \quad \le \sqrt{\frac{2L\Delta G}{T}}.$$

Proof. Descent lemma: $f(\mathbf{x}_{t+1}) \le f(\mathbf{x}_t) - \eta\nabla f(\mathbf{x}_t)^\top\mathbf{g}_t + \frac{L\eta^2}{2}\|\mathbf{g}_t\|^2$. Conditional expectation gives $\mathbb{E}_t f(\mathbf{x}_{t+1}) \le f(\mathbf{x}_t) - \eta\|\nabla f(\mathbf{x}_t)\|^2 + \frac{L\eta^2}{2}G$. Take total expectation (tower property), sum over $t < T$ (telescoping), use $\mathbb{E}f(\mathbf{x}_T) \ge f^\star$ and divide by ηT. Finally $\frac{a}{\eta} + b\eta$ is minimized at $\eta = \sqrt{a/b}$ with value $2\sqrt{ab}$. ■

The noise term forces $\eta \sim T^{-1/2}$, slower than GD's T^{-1}. Since mini-batches of size B divide the noise variance by B, larger batches reduce G and permit larger steps.

10 Momentum

Motivation. Gradient descent zigzags in narrow valleys. Momentum averages past gradients so that oscillating components cancel and consistent ones accumulate; Nesterov's variant evaluates the gradient at a look-ahead point.

10.1 Heavy-ball momentum

Definition 6.20: momentum

$\mathbf{v}_{t+1} = \beta\mathbf{v}_t - \eta\nabla f(\mathbf{x}_t)$, $\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_{t+1}$, $\mathbf{v}_0 = \mathbf{0}$, $0 \le \beta < 1$.

Unrolling, $\mathbf{v}_{t+1} = -\eta\sum_{k=0}^{t}\beta^{t-k}\nabla f(\mathbf{x}_k)$: an exponentially weighted sum of past gradients (a velocity). For a constant gradient $\mathbf{g}$, $\|\mathbf{v}_t\| \to \eta\|\mathbf{g}\|/(1-\beta)$, an effective step $1/(1-\beta)$ times larger ($10\times$ for $\beta = 0.9$). Persistent directions accumulate; oscillating directions cancel, which reduces zig-zagging in narrow valleys. Physically the gradient acts as a force and $1-\beta$ as friction: in the buffer form $\mathbf{m}_{t+1} = \mathbf{g}_t + \beta\mathbf{m}_t$ one has $\mathbf{m}_{t+1} - \mathbf{m}_t = \mathbf{g}_t - (1-\beta)\mathbf{m}_t$.

Theorem 6.21: momentum on a quadratic

For $f(x) = \frac{h}{2}x^2$, $x_{t+1} = (1+\beta-\eta h)x_t - \beta x_{t-1}$; the iteration converges iff both roots of $r^2 - (1+\beta-\eta h)r + \beta = 0$ lie in the open unit disk, which for $0 \le \beta < 1$ holds iff $0 < \eta h < 2(1+\beta)$. Complex roots (modulus $\sqrt{\beta}$) mean damped oscillation.

Proof. Substitute $\mathbf{v}_{t+1} = x_{t+1} - x_t$ and $\mathbf{v}_t = x_t - x_{t-1}$ into the velocity update. For the root condition use the Jury/Schur criterion for $r^2 + a_1 r + a_0$: $|a_0| < 1$, $1 + a_1 + a_0 > 0$, $1 - a_1 + a_0 > 0$, which here read $\beta < 1$, $\eta h > 0$, $\eta h < 2(1+\beta)$. ■

10.2 Nesterov momentum

Nesterov evaluates the gradient at the look-ahead point:

$$\mathbf{y}_t = \mathbf{x}_t + \beta \mathbf{v}_t, \qquad \mathbf{v}_{t+1} = \beta \mathbf{v}_t - \eta \nabla f(\mathbf{y}_t), \qquad \mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{v}_{t+1}.$$

Example 6.22: a numerical illustration

$f = x^2/2$, $x_0 = 1$, $\eta = 0.1$, $\beta = 0.9$. Both methods give $v_1 = -0.1$, $x_1 = 0.9$. Momentum: $v_2 = 0.9(-0.1) - 0.1(0.9) = -0.18$, $x_2 = 0.72$. Nesterov: $y_1 = 0.81$, $v_2 = -0.09 - 0.081 = -0.171$, $x_2 = 0.729$: it measures the slope where it is about to go and brakes earlier (not necessarily closer to the optimum at every step).

Theorem 6.23: two forms of Nesterov

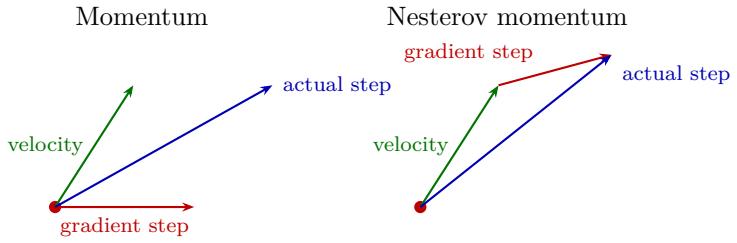
Form I: $\psi^{k+1} = \theta^k - \alpha g^k$, $\theta^{k+1} = \psi^{k+1} + \beta(\psi^{k+1} - \psi^k)$, $\psi^0 = \theta^0$. Form II (PyTorch `nesterov=True`): $m^{k+1} = g^k + \beta m^k$, $v^{k+1} = \beta m^{k+1} + g^k$, $\theta^{k+1} = \theta^k - \alpha v^{k+1}$, $m^0 = 0$. Here g^k is evaluated at θ^k. For the same θ^0 and $g^0, g^1, \ldots$ both produce the same $\theta^1, \theta^2, \ldots$

Proof. *Both satisfy one recursion.* Form I with $\psi^{k+1} = \theta^k - \alpha g^k$ and $\psi^k = \theta^{k-1} - \alpha g^{k-1}$ gives, for $k \geq 1$,

$$\theta^{k+1} = \theta^k - \alpha g^k + \beta(\theta^k - \theta^{k-1}) - \alpha\beta(g^k - g^{k-1}).$$

Form II: $\theta^{k+1} = \theta^k - \alpha g^k - \beta(\alpha g^k + \alpha\beta m^k)$, and from the previous step $\alpha\beta m^k = \alpha v^k - \alpha g^{k-1} = \theta^{k-1} - \theta^k - \alpha g^{k-1}$; substituting gives the same recursion. Both start with $\theta^1 = \theta^0 - \alpha(1+\beta)g^0$. A second-order recursion with the same two initial values has the same solution (induction). *Alternative.* Define $\psi^k := \theta^k + \alpha\beta m^k$ in Form II. Then $\psi^0 = \theta^0$, $\psi^{k+1} = \theta^k - \alpha g^k$ and $\psi^{k+1} - \psi^k = -\alpha m^{k+1}$, from which Form I's second equation follows. ■

Interpretation: ψ is the actual position (often written x_t), θ the look-ahead point $x_t + \beta v_t$ at which gradients are taken and which PyTorch stores as the parameter; m is the velocity of ψ scaled by $-1/\alpha$. Both converge to the same limit since $\theta^k - \psi^k = \beta(\psi^k - \psi^{k-1}) \to 0$.



Momentum adds the gradient step taken at the current point to the velocity; Nesterov evaluates the gradient after first moving along the velocity.

11 Adaptive Methods

Motivation. Different parameters often need different step sizes. AdaGrad, RMSProp and Adam rescale each coordinate by a running estimate of its gradient magnitude; this section derives their update rules and the bias correction in Adam.

All operations below are coordinatewise.

Definition 6.24: AdaGrad, RMSProp, Adam

AdaGrad: $\mathbf{r}_t = \mathbf{r}_{t-1} + \mathbf{g}_t \odot \mathbf{g}_t$, $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta\, \mathbf{g}_t \oslash (\sqrt{\mathbf{r}_t} + \epsilon)$.
RMSProp: $\mathbf{r}_t = \rho \mathbf{r}_{t-1} + (1-\rho)\mathbf{g}_t \odot \mathbf{g}_t$, same update.
Adam: $\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t$, $\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t \odot \mathbf{g}_t$, $\hat{\mathbf{m}}_t = \mathbf{m}_t/(1-\beta_1^t)$, $\hat{\mathbf{v}}_t = \mathbf{v}_t/(1-\beta_2^t)$, $\mathbf{x}_t = \mathbf{x}_{t-1} - \eta\, \hat{\mathbf{m}}_t \oslash (\sqrt{\hat{\mathbf{v}}_t} + \epsilon)$. Typical $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 10^{-3}$.

Remark 6.25: what each piece does

AdaGrad's accumulator grows forever, so its effective rate $\eta/\sqrt{r_t}$ shrinks (for a constant gradient g the step is $\eta/\sqrt{t}$). RMSProp forgets old gradients; for constant $g = 2$, $\rho = 0.9$, $\eta = 0.1$, $r_1 = 0.4$, $r_2 = 0.76$, steps $0.316, 0.229$, tending to 0.1. Adam = momentum's numerator (direction) + RMSProp's denominator $\sqrt{\mathbf{v}}$ (per-coordinate scale) + bias correction. (Nesterov plus Adam is NAdam.)

11.1 Exponential moving averages and bias correction

Theorem 6.26: EMA of iid gradients

If $g_0, g_1, \ldots$ are iid with mean μ and variance s^2, and $m_{k+1} = \beta m_k + (1-\beta) g_k$, $m_0 = 0$, then

$$\mathbb{E} m_k = (1-\beta^k)\mu, \qquad \operatorname{Var} m_k = s^2 \frac{1-\beta}{1+\beta}(1-\beta^{2k}),$$

with limits μ and $s^2 \frac{1-\beta}{1+\beta}$. The corrected $\hat{m}_k = m_k/(1-\beta^k)$ is unbiased with variance $s^2 \frac{(1-\beta)(1+\beta^k)}{(1+\beta)(1-\beta^k)}$ (equal to s^2 at $k = 1$).

Proof. m_k depends only on $g_0, \ldots, g_{k-1}$, so it is independent of g_k. Mean by induction: $\beta(1-\beta^k)\mu + (1-\beta)\mu = (1-\beta^{k+1})\mu$. Variance: $\operatorname{Var} m_{k+1} = \beta^2 \operatorname{Var} m_k + (1-\beta)^2 s^2$; writing $(1-\beta)^2 = \frac{1-\beta}{1+\beta}(1-\beta^2)$, the induction step is $\beta^2(1-\beta^{2k}) + (1-\beta^2) = 1 - \beta^{2k+2}$. Correction: divide by $(1-\beta^k)^2$ and use $1-\beta^{2k} = (1-\beta^k)(1+\beta^k)$. ∎

Without correction Adam's first step is $\frac{1-\beta_1}{\sqrt{1-\beta_2}} \operatorname{sign}(g_1)\eta \approx 3.16\,\eta$; with correction $\hat{m}_1 = g_1$, $\hat{v}_1 = g_1^2$ and the step is exactly $\eta \operatorname{sign}(g_1)$. The correction removes the zero-initialization bias at the price of a larger early variance.

11.2 Special cases and invariances

Theorem 6.27: Adam contains RMSProp and signSGD

For Adam without bias correction: $\beta_1 = 0$ gives RMSProp with $\rho = \beta_2$; $\beta_1 = \beta_2 = 0$, $\epsilon = 0$ give signSGD $\theta_{k+1} = \theta_k - \eta \operatorname{sign}(g_k)$ (if no coordinate of g_k is 0). With $\epsilon = 0$, Adam's iterates are unchanged when the loss is multiplied by $c > 0$, while SGD's step scales by c.

Proof. $\beta_1 = 0$ makes $m_{k+1} = g_k$. With $\beta_2 = 0$ also, $\sqrt{v_{k+1}} = |g_k|$. Scaling $g \mapsto cg$ scales m by c and $\sqrt{v}$ by c. ∎

11.3 Weight decay is not ℓ_2 regularization

For plain SGD, ℓ_2 regularization $\theta - \eta(g + \lambda\theta) = (1-\eta\lambda)\theta - \eta g$ is the same as weight decay. With momentum it is not:

Theorem 6.28: no matching exists

SGD with momentum $v_{k+1} = g_k + \beta v_k$, $\theta_{k+1} = \theta_k - \alpha v_{k+1}$ and $\alpha, \beta > 0$. Weight decay: $\theta_{k+1} = \theta_k - \alpha v_{k+1} - \mu\theta_k$. ℓ_2: $v_{k+1} = (g_k + \lambda\theta_k) + \beta v_k$. No choice of $\mu, \lambda > 0$ makes the two methods equivalent.

Proof. Take $g_k \equiv 0$, $v_0 = 0$, $\theta_0 \neq 0$. Weight decay: $v_k = 0$ and $\theta_k = (1-\mu)^k\theta_0$. ℓ_2: $v_1 = \lambda\theta_0$, $\theta_1 = (1-\alpha\lambda)\theta_0$, $v_2 = \lambda(1-\alpha\lambda+\beta)\theta_0$, $\theta_2 = [(1-\alpha\lambda)^2 - \alpha\beta\lambda]\theta_0$. Matching θ_1 forces $\mu = \alpha\lambda$, and then matching θ_2 forces $\alpha\beta\lambda = 0$, a contradiction. (Indeed the ℓ_2 sequence is geometric for no ratio.) ■

In Adam the ℓ_2 gradient $\lambda\theta$ is divided by its own RMS, so every weight moves by about η regardless of its size ($\theta_1 = \theta_0 - \eta\,\mathrm{sign}(\theta_0)$ when $g \equiv 0$), losing proportional shrinkage; AdamW applies the decay $(1-\eta\lambda)\theta$ outside the adaptive scaling.